

Hanping Xu

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PERSONAL DATA

CITIZENSHIP AND YEAR OF BIRTH: Chinese | 1996
EMAIL: xuhanping@u.nus.edu

EDUCATION

2018 - 2023 Ph.D. in MATHEMATICAL ECONOMICS, **National University of Singapore (NUS)**, Singapore
Advisor: Prof. Yeneng Sun GPA: 4.89/5.0

2014 - 2018 B.S. in SCIENCE with Honors Degree, **Sun Yat-sen University (SYSU)**, Guangzhou, China
Major: Statistics

WORKING EXPERIENCE

2024.3 - Present Postdoctoral Research Fellow, **The Chinese University of Hong Kong (CUHK)**, Hong Kong

2023.1 - 2023.12 Research Assistant, **The Chinese University of Hong Kong (CUHK)**, Hong Kong, China

2022.8 - 2022.12 Student Research Assistant, **National University of Singapore (NUS)**, Singapore

RESEARCH INTERESTS

Game Theory, Microeconomic Theory, Mathematical Economics

PUBLICATION

[Pareto-undominated and socially-maximal Nash equilibria with coarser traits](#) (with [Bin Wu](#)), Economics Letters, 215(2022) : 110464.

WORKING PAPERS

[Monotone Perfection](#)

with [Wei He](#) and [Yeneng Sun](#)

Abstract: Monotone equilibria may be undesirable in Bayesian games in the sense that players adopt weakly dominated strategies. To account for the possibility that the players might choose unintended strategies through a trembling hand, we propose an equilibrium refinement called “perfect monotone equilibrium.” This notion strengthens the notion of monotone equilibrium in the sense that it satisfies the important property of admissibility in Bayesian games with finitely many actions, and the property of limit undominatedness in Bayesian games with infinitely many actions.

In a general class of Bayesian games where each player’s action set is a sublattice of multi-dimensional

Euclidean space and players' types are also multi-dimensional, a perfect monotone equilibrium is shown to exist under the supermodularity and increasing differences conditions. These conditions model the scenarios in which, informally, players' payoffs have complementarity in own actions and monotone incremental returns in own types. We demonstrate that the increasing differences condition is sharp by providing a two-player game that satisfies the widely adopted single crossing condition in the literature, but does not possess any perfect monotone equilibrium. To show the usefulness of our result in the setting with discontinuous payoffs, we provide various illustrative applications, including first-price auctions, all-pay auctions, and Bertrand competitions. Our result can be further extended to the setting with more general action spaces and type spaces.

The equilibrium property of obvious strategy profiles in games with many players

with [Enxian Chen](#) and [Bin Wu](#)

Abstract: This paper studies the equilibrium properties of the “obvious strategy profile” in large finite-player games. Each player in such a strategy profile simply adopts a randomized strategy as she would have used in a symmetric equilibrium of an idealized large game. We show that, under a continuity assumption, (i) obvious strategy profiles constitute a convergent sequence of approximate symmetric equilibria as the number of players tends to infinity, and (ii) realizations of such strategy profiles also form a convergent sequence of (pure strategy) approximate equilibria with probability approaching one. Our findings offer a solution that is easily implemented without coordination issues and is asymptotically optimal for players in large finite games. Additionally, we present a convergence result for approximate symmetric equilibria.

Equilibrium convergence in large games

with [Enxian Chen](#) and [Bin Wu](#)

Abstract: This paper presents a general closed graph property for (randomized strategy) Nash equilibrium correspondence in large games. In particular, we show that for any large game with a convergent sequence of finite-player games, the limit of any convergent sequence of Nash equilibria of the corresponding finite-player games can be induced by a Nash equilibrium of the large game. Such a result goes beyond earlier results on the closed graph property for pure strategy Nash equilibrium correspondence in large games in multiple aspects. An application on equilibrium selection in large games is also presented.

Does Randomization Matter in Sequential Games?

with [Enxian Chen](#), [Wei He](#), and [Yeneng Sun](#)

Abstract: We provide an example to demonstrate that randomization matters in sequential games in the sense that players may obtain higher equilibrium payoffs by playing mixed strategies. In contrast, we present a fundamental property for the classic two-player zero-sum sequential games (such as chess and go), showing that realizations of mixed-strategy subgame perfect equilibria (SPE) induce pure-strategy SPEs with the same payoffs. As an application, we show a new existence result on pure-strategy SPE in such games. A universal SPE that induces all the pure-strategy SPE is constructed. Finally, we prove that the above fundamental property holds generically in sequential games.

Games with incomplete information: a general purification result

with [Wei He](#) and [Yeneng Sun](#)

Abstract: We present a new purification result for Bayesian games with countably many actions, interdependent payoffs and correlated types. It is shown that the condition of coarser inter-player information characterizes the existence of purification, and also the existence of pure strategy equilibrium in these games. We demonstrate that the condition of countably many actions is tight for the purification result and pure strategy equilibrium existence. To prove the results for Bayesian games, we provide a general purification principle, which covers various earlier results as special cases.

Large games with coarser traits and countable actions

Abstract: We show that the coarser traits condition is both necessary and sufficient for the idealized limit property of large games with traits and countable actions. Meanwhile, we also show that the coarser traits condition is both necessary and sufficient for the existence of pure strategy Pareto-undominated

socially-maximal Nash equilibria in large games with traits and countable actions. We demonstrate that the condition of countably many actions is tight.

TEACHING EXPERIENCE

<i>National University of Singapore</i>	MA4264: GAME THEORY	2021/2022 SEM 2
	MA4264: GAME THEORY	2020/2021 SEM 2
	MA3238: STOCHASTIC PROCESSES I	2019/2020 SEM 2
	MA2216: PROBABILITY	2019/2020 SEM 1
<i>Teaching evaluation score</i>	4.3/5.0 for Game Theory	4.1/5.0 for Stochastic Process I

PRESENTATIONS

2024 THE SEVENTH WORLD CONGRESS OF THE GAME THEORY SOCIETY
Presenter for “Monotone Perfection”

2024 CHINA MEETING ON GAME THEORY AND ITS APPLICATIONS
Presenter for “Obvious approximate symmetric equilibrium in games with many players”

2022 SOCIETY FOR THE ADVANCEMENT OF ECONOMIC THEORY (SAET) Conference
Presenter for “Obvious approximate symmetric equilibrium in games with many players”

2021 SOCIETY FOR THE ADVANCEMENT OF ECONOMIC THEORY (SAET) Conference
Presenter for “Games with incomplete information: a general purification result”

SELECTED SCHOLARSHIPS, HONORS AND AWARDS

PART-TIME TEACHING ASSISTANT AWARD, Faculty of Science, NUS	2022
RESEARCH SCHOLARSHIPS, NUS	2019, 2020, 2021, 2022
TOP GRADUATE TUTOR, Department of Mathematics, NUS	2021
OUTSTANDING GRADUATE AWARD, Sun Yat-sen University	2018
NATIONAL SCHOLARSHIPS, China	2015, 2016, 2017
THE FIRST PRIZE SCHOLARSHIPS, Sun Yat-sen University	2015, 2016, 2017

REVIEWER FOR JOURNAL

Economic Theory, Journal of Mathematical Economics

REFERENCES

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